

ORIGINAL ARTICLE

Diversity of *Vibrio parahaemolyticus* in marine fishes of Bangladesh

S. Ali¹, M. Hossain¹, A.B. Azad³, A.B. Siddique¹, M. Moniruzzaman¹, M.A. Ahmed¹, M.B. Amin², M.S. Islam¹, M.M. Rahman⁴, D. Mondal^{1,2} and Z.H. Mahmud¹ 

1 Laboratory of Environmental Health, Laboratory Sciences and Services Division, International Centre for Diarrhoeal Disease Research, Bangladesh, Dhaka, Bangladesh

2 Laboratory of Food Safety and One Health, Laboratory Sciences and Services Division, International Centre for Diarrhoeal Disease Research, Bangladesh, Dhaka, Bangladesh

3 Department of Botany, University of Dhaka, Dhaka, Bangladesh

4 Department of Microbiology, University of Dhaka, Dhaka, Bangladesh

Keywords

Bangladesh, biofilm, diversity, ERIC-PCR, marine fish, serology, *Vibrio parahaemolyticus*.

Correspondence

Zahid Hayat Mahmud, Laboratory of Environmental Health, International Center for Diarrhoeal Disease Research, Bangladesh (icddr,b), 68, Shaheed Tajuddin Ahmed Sarani, Mohakhali, Dhaka 1212, Bangladesh. E-mail: zhmahmud@icddr.org

2021/0013: received 4 January 2021, revised 4 March 2021 and accepted 26 March 2021

doi:10.1111/jam.15093

Abstract

Aims: To determine the occurrence, diversity, antibiotic resistance and biofilm formation of *Vibrio parahaemolyticus* isolated from marine fishes in Bangladesh.

Methods and Results: A total of 80 marine fishes were obtained from the local markets and examined for the presence of *V. parahaemolyticus*. All the isolated *V. parahaemolyticus* were characterized for the presence of virulence markers, thermostable direct hemolysin (TDH) or thermostable direct hemolysin related hemolysin (TRH). Isolates were serotyped and further characterized by enterobacterial repetitive intergenic consensus sequence PCR (ERIC-PCR) typing to analyse the genetic diversity. Moreover, biofilm formation and antibiotic resistance patterns were also determined. About 63.75% (51/80) of the tested marine fishes were contaminated with *V. parahaemolyticus*. From the contaminated fishes, 71 representatives *V. parahaemolyticus* were isolated and none of them harboured *tdh* and *trh* virulence genes. Nine different O-groups and seven different K-types were found by serological analysis and the dominant serotype was O5:KUT. In ERIC-PCR analysis, eight clusters (A–H) were found and the most common pattern was A (46.5%). All of the isolates were resistant to ampicillin and 78.9% of isolates were resistant to streptomycin. The highest biofilm formation was found at 37°C compared to 25°C and 4°C.

Conclusion: Diverse *V. parahaemolyticus* are present in marine fishes in the local market of Bangladesh with antibiotic-resistant properties and biofilm formation capacity.

Significance and Impact of the Study: The widespread prevalence of diverse *V. parahaemolyticus* in marine fishes is an issue of serious concern, and it entails careful monitoring to ascertain the safety of seafood consumers.

Introduction

Vibrio parahaemolyticus is a well-known Gram-negative halophilic bacterium, widely present in salt water, estuarine and seaside aquatic environments including lives, such as fish, shrimp and zooplankton (Joseph *et al.* 1982;

Baffone *et al.* 2006; Mahmud *et al.* 2006; Ceccarelli *et al.* 2013; Letchumanan *et al.* 2014). This bacterium is known globally as a major source of foodborne disease and, due to the ingestion of undercooked and untreated seafood, has also been responsible for acute human gastroenteritis with further complications such as septicemia (Qadri

et al. 2005; Letchumanan *et al.* 2014). In Asia, the United States (Daniels *et al.* 2000) and some European countries (Martinez-Urtaza *et al.* 2005; Quilici *et al.* 2005), epidemiological reports suggest that *V. parahaemolyticus* is responsible for foodborne associated outbreaks. Bangladesh has also recorded a few incidences of *V. parahaemolyticus* (Bhuiyan *et al.* 2002; Islam *et al.* 2004; Alam *et al.* 2009; Siddique *et al.* 2021). However, concerns about the growing incidence of *V. parahaemolyticus* in Bangladesh have been posed by the rise in the production and consumption of sea fish and the number of susceptible individuals.

The exact mechanism of how *V. parahaemolyticus* infects humans has yet to be completely identified. The most important virulence factors for this organism are thermostable direct hemolysin (TDH) and TDH-related hemolysin (TRH), and any or both of these virulence factors are responsible for the disease (Honda *et al.* 1989; Alam *et al.* 2009; Wang *et al.* 2015). Both TDH and TRH have the same hemolytic mechanisms *in vitro* to break down human erythrocytes. Both of these toxins have other virulence properties like enterotoxin activity, cardio-toxicity, and cytotoxicity (Tang *et al.* 1995; Matsuda *et al.* 2010). However, only less than 2 % of *V. parahaemolyticus* isolates of environmental origins were tested positive for the virulence factors (Okuda *et al.* 1997; Drake *et al.* 2007). Most of these isolates are considered nonpathogenic and are particularly susceptible to most medically used antibiotics. However, for therapy and prophylaxis, antibiotics are being used aggressively (Jerbi *et al.* 2011), leading to the development of resistant strains (Tendencia & de la Peña 2001). This poses a potential threat to humans because these resistant bacteria can transmit directly to the human body through food and other human pathogens can adopt these resistance genes via mobile genetic elements (Duran & Marshall 2005; Guglielmetti *et al.* 2009).

There are currently 13 O antigens and 71 K antigens of *V. parahaemolyticus*. The O3: K6 pandemic strain, the most commonly identified from clinical isolates, has been known since 1996 (Honda *et al.* 2008). However, it is not yet known how the latest pandemic arose from this particular serotype. Several molecular typing methods, namely pulsed-field gel electrophoresis, ERIC PCR, ribotyping, random amplified polymorphic DNA and multi-locus sequence typing, were applied to the characterization of *V. parahaemolyticus* (Marshall *et al.* 1999; Chen *et al.* 2012). ERIC PCR was recommended to apply in the characterization of *V. parahaemolyticus* based on their quick, labour-saving, high discriminative and reproducible capacities.

In extreme condition, bacteria form a biofilm which helps them to survive that harsh condition (Guglielmetti

et al. 2009). Biofilms are complex surface-attached communities of single or mixed types of organisms where multiple cells form a solid complex, self-assembled, three-dimensional extracellular polymer (EPS) containing exopolysaccharides, proteins, extracellular DNA (eDNA), etc. (Hara-Kudo *et al.* 2001; Honda *et al.* 2008). EPS keeps the cells attached in biofilm, enables strong interactions, including intercellular contact, horizontal transmission of genes among the cells and helps to develop synergistic micro-consortia (Bej *et al.* 1999). *V. parahaemolyticus* has the ability to develop biofilms on various living or nonliving surfaces, such as oysters, stainless steel (SS), etc. (Krumperman 1983). In addition, on average, the pathogenic *V. parahaemolyticus* can develop more biofilms than nonpathogenic *V. parahaemolyticus* at all temperatures measured (O'Toole 2011).

Marine fish contribute as one of the major protein sources in Bangladesh. Over the last decade, the cultivation, distribution and consumption of marine fish have been continuously increased. However, recent outbreaks of O3: K6 causing foodborne diseases on the subcontinent call for scientific attention to the environmental monitoring of marine waters and marine fish in Bangladesh. Though *V. parahaemolyticus* is a marine organism, its pandemic spread is very little understood. Consequently, this study was piloted to distinguish and examine the occurrence, serological diversity, genotypic characteristics, pathogenic as well as the biofilm-forming capacity of *V. parahaemolyticus* strains present in marine fishes collected from Bangladeshi local markets.

Materials and Methods

Sample collection and processing

A total of 80 marine fishes of 12 different species were purchased from the different local markets in Dhaka, Khulna and Cox's Bazar (Table 1). They were collected throughout the year on the basis of their availability and their distribution throughout the year has been shown in Table S1. Every fish sample was collected in a sterile Whirl-Pak zipper bag (Fisherbrand™, Waltham, MA, USA.) from fish markets, then labelled and kept in a cool box and shipped to the Laboratory of Environmental Health, maintaining the temperature 4–10°C and the samples were processed just after receiving. At first, the samples were washed with sterile 2% NaCl solution in the laboratory to remove surface microorganisms, dirt, sand and slime. Then the flesh, gill and gut were divided aseptically according to the recommended procedure (Kaysner & DePaola 2004).

Table 1 Fish samples used in this study

Serial no.	Local name	Scientific name	No. of tested fishes	No. of contaminated fishes	Contamination %
1	Khaira	<i>Sardinella fimbriata</i>	6	4	66.7
2	Chapila	<i>Gudusia chapra</i>	8	5	62.5
3	Poa, Poma	<i>Argyrosomus amoyensis</i>	6	4	66.7
4	Loittia	<i>Harpodon nehereus</i>	8	5	62.5
5	Bom Maitta, Tuna	<i>Euthynnus affinis</i>	6	4	66.7
6	Rup Chanda	<i>Pampus chinensis</i>	6	3	50.0
7	Kata Phasa	<i>Stolephorus tri</i>	8	6	75.0
8	Ilish	<i>Hilsa ilisha</i>	8	6	75.0
9	Tuitta, Banded Neddle fish	<i>Strongylura leiura</i>	6	2	33.3
10	Gutra Ilish	<i>Hilsa kelee</i>	6	5	83.3
11	Apuia	<i>Osteogeneiosus militaris</i>	6	4	66.7
12	Bata	<i>Liza subviridis</i>	6	3	50.0
		Total	80	51	63.8

Isolation of *V. parahaemolyticus*

Flesh, gill, and gut from 80 marine fishes were separated aseptically and analysed for the presence of *V. parahaemolyticus*. Then, the separated parts (25 g or less) were homogenized adequately with nine times alkaline peptone water containing 3% NaCl and kept at 37°C overnight for culture enrichment. After incubation, a loopful culture broth was streaked onto thiosulfate citrate bile salt sucrose (TCBS) agar (BD, NJ) and incubated for 18–24 h at 37°C. Colonies that appeared green on TCBS agar plate were subsequently screened and subcultured again on plates of CHROMagar Vibrio (CV) agar (CHROMagar, Paris, France). Violet colonies were deemed to be *V. parahaemolyticus* on CV agar plates (Hara-Kudo *et al.* 2001). Using the API 20E method (bioMérieux, Marcy l'Etoile, France), these isolates were further confirmed. Stock cultures were preserved with 30 % glycerol at –80 °C in Luria Bertani (LB) broth (BD, NJ).

Extraction of Chromosomal DNA

One isolated colony of *V. parahaemolyticus* from Gelatin Agar (GA) plate (BD, NJ) was kept for overnight incubation at 37°C in a shaker incubator after inoculating into a 5 ml LB broth containing 3% NaCl. Following incubation, 1.0 ml of culture was centrifuged at 5 000 *g* for 4 min and the supernatant was removed. The collected cell pellet was resuspended in the same volume of autoclaved distilled water and thoroughly mixed. The suspension was then boiled for 10 min at 100°C and put immediately in the ice for 15 min. After that, the suspension was centrifuged at 2000 *g* for 10 min. Finally, 100 µl of the supernatant was collected and preserved at –20°C until further use.

Multiplex PCR amplification

A multiplex PCR was carried out following the procedure previously described with minor modifications (Bej *et al.* 1999). Briefly, the PCR was conducted using 12.5 µl Go Taq 2X Green Master Mix (Promega, CA), 0.5 µg of genomic DNA, 0.5 mmol l^{–1} of each of the oligonucleotide primers for *tlh*, *tdh*, and *trh*, and an appropriate volume of sterile nuclease-free water to make a 25 µl reaction mixture. The PCR was run in a thermal cycler (Bio-Rad, Singapore), maintaining the following protocol: initial denaturation for 5 min at 95°C, subsequently 35 cycles of amplification. In every cycle, PCR condition was set for denaturation at 95°C, primer annealing at 58°C, and primer extension at 72°C for 1 min each. After the amplification cycles, samples were held for 10 min at 72°C to allow the partially synthesized DNA extension. Electrophoresis was carried out to separate the amplified products in the Midori Green Advance stained 1.5% agarose gel. Finally, the gel was visualized and imaged under a UV transilluminator (Bio-Rad, Singapore City, Singapore).

Serogrouping

The serotyping of the isolated samples was performed by slide agglutination test using the antiserum test kit (Denka Seiken, Tokyo, Japan) for *V. parahaemolyticus* following the instructions of the manufacturer. In short, the bacteria were first subcultured on LB agar plate containing 3% NaCl and incubated overnight at 37°C. A loopful of fresh inoculum was suspended in autoclaved normal saline after incubation. Then, half of the cell suspension was boiled for 2 h and used for serotyping on the basis of O antigen. The rest was used for K antigen-based serotyping.

Antibiotic susceptibility testing

Kirby Bauer disc diffusion method was followed to check the susceptibility of the *V. parahaemolyticus* isolates to the different classes of antibiotics, maintaining CLSI guidelines (Hindler & Jorgensen 2011). The 14 antibiotics discs (Oxoid) from the different groups were used: ampicillin (10 µg), piperacillin/tazobactam (110 µg), cefotaxime (30 µg), cefuroxime (30 µg), imipenem (10 µg), meropenem (10 µg), gentamicin (10 µg), amikacin (30 µg), streptomycin (10 µg), tetracycline (30 µg), ciprofloxacin (5 µg), nalidixic acid (30 µg), trimethoprim-sulfamethoxazole (25 µg), and chloramphenicol (30 µg). *Escherichia coli* ATCC 25922 strain was used as a standard organism in this study. Three individual experiments were conducted to obtain the result. The average value of inhibition zone (mm) of each strain against different antibiotics is presented in Table S3 and strain were categorized as resistant, intermediate and sensitive according to CLSI guidelines (Table S4) (Patel *et al.* 2015). The multiple antibiotic resistance (MAR) index was estimated on the basis of the ratio between the number of antibiotic-resistant isolates and the number of screened antibiotics for the isolates. Multidrug resistance (MDR) was characterized when an isolate is resistant to at least one antimicrobial agent of three or more classes of antibiotics (Krumperman 1983).

Biofilm formation study

The biofilm formation experiment of *V. parahaemolyticus* at three separate temperatures (4, 25 and 37°C) was carried out in accordance with the procedure defined by O'Toole with a small modification (O'Toole 2011). First, the bacterial culture was grown in LB broth media for 24 h in the shaker incubator, and then it was diluted (1 : 50) in fresh LB broth. On 96-well microtiter plates, 100 µl of each diluted culture was taken and kept for overnight incubation at 4, 25 and 37°C. The culture was discarded after incubation and washed in a water tub. In each well of the microtiter plates, 120 µl of a 0.1% solution of crystal violet was gently added and then those were kept at room temperature for 10 min. After that, the plates were rinsed three times and immersed in a water tub and dried at room temperature for 15 min. Finally, 125 µl of 95% (v/v) ethanol was applied to each well and absorption was quantified at 590 nm in EPOCH2 microplate reader (BioTek, Winooski, VT). Un-inoculated LB was used as a negative control in this experiment. Each isolate was tested for three wells, and the mean OD was determined. Optical density cut-off (ODc) was assigned as an average OD of negative controls + (3 × standard deviation (SD) of negative controls). Isolate with OD ≤ ODc was categorized as nonbiofilm producer. Meanwhile, the isolate was

categorized as biofilm producer consisting of weak biofilm producer if $ODc < OD \leq 2 \times ODc$; moderate biofilm $2 \times ODc < OD \leq 4 \times ODc$; and strong biofilm producer if $OD > 4 \times ODc$ (Hassan *et al.* 2011). Three individual experiments were conducted and the data were represented as the mean ± standard deviation.

ERIC-PCR

Two primer sequences ERIC-1 (ATGTAAGCTCCTGGG-GATTAC) and ERIC-2 (AAGTAAGTGACTGGGGT-GAGCG) (Versalovic *et al.* 1991) were used for ERIC PCR to check the clonal association among the isolates. The conducted amplification conditions were as described by Rivera (Rivera *et al.* 1995). Through gel electrophoresis, 20 µl of amplified PCR products were loaded to separate the bands. In this experiment, 2% agarose gel was used, and the run was continued for 240 min at 60 V. Invitrogen 1 kb Plus Ladder (Fisher Scientific, Waltham, MA) was used as DNA markers and ethidium bromide (EtBr) (0.5 mg ml⁻¹) was used to stain the gel for visualization. Gels were imaged under a UV transilluminator (Bio-Rad) and images were analysed by diversity database finger-printing software, BioNumerics version 4.5 (Applied Maths, Kortrijk, Belgium) to observe the clonal diversity among the strains (Rivera *et al.* 1995). Later, the unweighted average pair group method (UPGMA) was followed to prepare a dendrogram.

Results

Prevalence of *V. parahaemolyticus* in marine fishes

In this study, a total of 80 marine fishes were examined and 51 (63.75%) were found as positive for the presence of *V. parahaemolyticus*. A total of 71 representative *V. parahaemolyticus* were isolated and confirmed by PCR targeting species-specific *tlh* gene, but none of them harboured the virulence genes *tdh* and/or *trh*. Contamination with *V. parahaemolyticus* varied with different parts of these fish samples (Table S2). The presence of *V. parahaemolyticus* was found highest in the gut followed by gill and lowest in the flesh of the marine fishes. The findings showed a year-round difference in the occurrence of *V. parahaemolyticus* in different months. The incidence was higher during the warmer months (April–October) compared to the colder months (November–March) (Fig. 1).

Susceptibility of *V. parahaemolyticus* isolates to antibiotics

In this study, we used a total of 14 antibiotics recommended by the Centre for Disease Control and Prevention

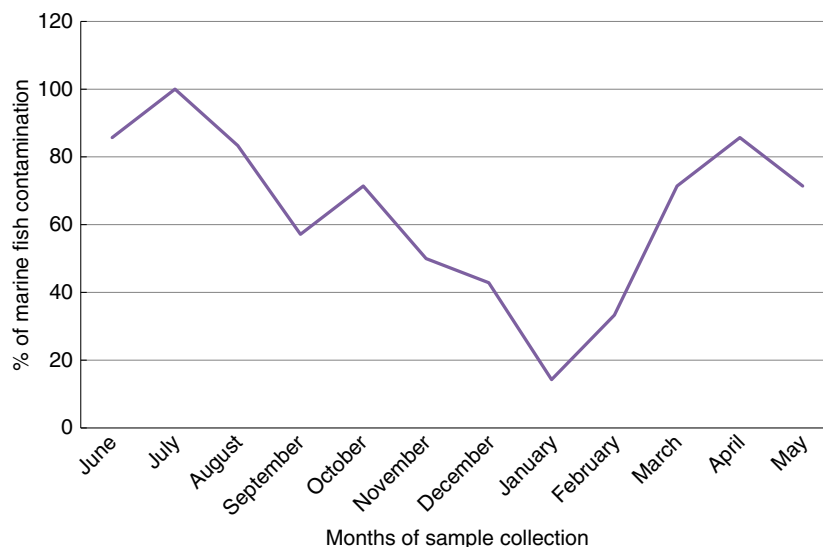


Figure 1 Monthly incidence rates of *V. parahaemolyticus* in marine fish samples in Bangladesh. The data of incidence rate were defined as (positive sample number/total sample number) $\times$ 100.

(CDC) to treat *Vibrio* associated infections. Interestingly, we found that all the 71 (100%) isolates showed resistance to ampicillin. In addition to ampicillin, we also found that isolates displayed a high resistant profile to the aminoglycosides class. A total of 56 (78.9%) isolates were found resistant to streptomycin, 21 (29.6%) to amikacin, and 5 (7%) to gentamicin. A high percentage (18.3%) of antibiotic resistance was also observed among the *V. parahaemolyticus* isolates to cefuroxime. However, high susceptibility to antibiotics including meropenem (100%), chloramphenicol (100%), imipenem (97.2%), nalidixic acid (88.7%), piperacillin/tazobactam (85.9%), trimethoprim-sulfamethoxazole (81.7%), cefotaxime (80.3%), ciprofloxacin (77.5%), and gentamicin (70.4%) was observed among the *V. parahaemolyticus* isolates (Table 2).

A high proportion (36.7%) of isolates showed a MAR index greater than 0.2 in the current analysis. The MAR index ranged from 0.07 to 0.37, where the highest MAR index (0.37) was found to five isolates (VP-13, VP-28, VP-42, VP-44 and VP-63), indicating resistance to five different antibiotics (Table 3). Approximately 2.6 and 26.7% of isolates were resistant to four and three different antibiotics, respectively, with a MAR index of 0.29 and 0.21. Here, VP denotes *V. parahaemolyticus*.

Diverse serotypes of *V. parahaemolyticus* present in marine fishes

During this study, all *V. parahaemolyticus* isolates were serotyped to show serological diversity and pandemic properties. Of the 71 isolates, 55 were categorized into 9 different O-groups (O1, O2, O3, O4, O5, O6, O9, O10 and O11) and 16 were untyped O-group. In our study,

the most frequently observed groups were O5 (29.58%), O10 (14.08%), O2 (11.27%), O11 (7.04%) and O1 (7.04%). Most of the isolates were K-typing untypable from 71 isolates given; only seven different K-types were determined by K-typing specific. The most common serotype was O5:KUT (with 19 isolates). Ten O10:KUT strains, seven O2:KUT, five O1:KUT, five O11:KUT, two O4:KUT and one each of O2:K25, O3:K7, O4:K49, O5:K17, O5:K32, O6:K3, O9:KUT and OUT:K53 were involved in other O: K serogroups (Table 4).

Biofilm formation varies with temperature

Biofilm development of *V. parahaemolyticus* isolates at three separate temperatures was presented in Fig. 2. About 84.6, 74.4, and 43.6% of the isolates produced biofilm at the three different temperatures measured (37°C, 25°C and 4°C, respectively). In our study, majority of the isolates (56.4%) could not produce biofilm at lower temperature (4°C) but rest of the isolates produced a weak or moderate biofilm. Biofilm development by *V. parahaemolyticus* isolates was dissimilar at 4°C compared to 25°C and 37°C. The most stable biofilm development of *V. parahaemolyticus* was observed at 37°C, compared to two other temperatures (4°C and 25°C).

ERIC-PCR confirms the genetic diversity of *V. parahaemolyticus*

All of the 71 isolates were ERIC-PCR typeable; comprising 4–11 bands from 100 bp to around 1600 bp. The ERIC image showed that bands with about 150 and 500 bp molecular sizes are common in most isolates. The

Table 2 Antibiotic resistance profiles of *V. parahaemolyticus* isolates from marine fishes

Antibiotic class	Antibiotics	Interpretive categories		
		Number (%) of S	Number (%) of I	Number (%) of R
Penicillin	Ampicillin (10 µg)	0 (0.0)	0 (0.0)	71 (100.0)
	Piperacillin/tazobactam (110 µg)	61 (85.9)	10 (14.1)	0 (0.0)
Cephalosporins	Cefotaxime (30 µg)	57 (80.3)	13 (18.3)	1 (1.4)
	Cefuroxime (30 µg)	19 (26.8)	39 (54.9)	13 (18.3)
Carbapenems	Imipenem (10 µg)	69 (97.2)	2 (2.8)	0 (0.0)
	Meropenem (10 µg)	71 (100.0)	0 (0.0)	0 (0.0)
Aminoglycosides	Gentamicin (10 µg)	50 (70.4)	16 (22.5)	5 (7.0)
	Amikacin (30 µg)	12 (16.9)	38 (53.5)	21 (29.6)
	Streptomycin (10 µg)	0 (0.0)	15 (21.1)	56 (78.9)
Tetracycline	Tetracyclin (30 µg)	66 (93.0)	3 (4.2)	2 (2.8)
Quinolones	Ciprofloxacin (5 µg)	55 (77.5)	16 (22.5)	0 (0.0)
	Nalidixic Acid (30 µg)	63 (88.7)	8 (11.3)	0 (0.0)
Sulfonamides	Trimethoprim-sulfamethoxazole (25 µg)	58 (81.7)	11 (15.5)	2 (2.8)
Phenicol	Cholramphenicol (30 µg)	71 (100.0)	0 (0.0)	0 (0.0)

Table 3 Multiple antibiotic resistance (MAR) index value of *V. parahaemolyticus* isolates from marine fish samples

MAR index	Antibiotic resistance profile	Isolate ID	Percentage of Isolates
0.37	AMP-CXM-AK-S-TE	VP-28	1 (1.3)
	AMP-CXM-CN-AK-S	VP-42, VP-44, VP-63	3 (4.23)
	AMP-CXM-CTX-AK-S	VP-13	1 (1.3)
0.29	AMP-CN-AK-S	VP-2	1 (1.3)
	AMP-CXM-AK-S	VP-41	1 (1.3)
0.21	AMP-AK-S	VP-3, VP-9, VP-12, VP-14, VP-16, VP-23, VP-24, VP-26, VP-27, VP-36, VP-49, VP-65	12 (16.0)
	AMP-CXM-S	VP-6, VP-19, VP-29, VP-51, VP-58	5 (7.0)
	AMP-S-TE	VP-7	1 (1.3)
	AMP-CN-AK	VP-54	1 (1.3)
0.14	AMP-S	VP-1, VP-4, VP-5, VP-8, VP-10, VP-11, VP-15, VP-18, VP-20, VP-21, VP-22, VP-25, VP-30, VP-31, VP-32, VP-34, VP-35, VP-37, VP-38, VP-39, VP-40, VP-45, VP-46, VP-47, VP-48, VP-50, VP-64, VP-66, VP-67, VP-69, VP-70	31 (70.4)
	AMP-CXM	VP-33, VP-61	2 (2.82)
	AMP-SXT	VP-55, VP-62	2 (2.82)
	AMP-AK	VP-71	1 (1.3)
	AMP	VP-17, VP-43, VP-52, VP-53, VP-56, VP-57, VP-59, VP-60, VP-68	9 (12.68)

AMP, Ampicillin; CXM, Cefuroxime; AK, Amikacin; S, Streptomycin; TE, Tetracycline; CN, Gentamicin; CTX, Cefotaxime; SXT, Trimethoprim-sulfamethoxazole.

ERIC-PCR fingerprints distributed the isolates into eight clusters at a 55% cut-off coefficient of Dice similarity, among which cluster A was the predominant (46.48%) (Fig. 3). Clusters B and D contained only one strain each. Clusters A, E, F, G, and H represented strains from all the three locations from where samples were collected, whereas clusters C constituted only a particular location. Grouping of the isolates based on antibiotic resistance patterns and serotypes was not observed in ERIC-PCR fingerprint analysis.

Discussion

V. parahaemolyticus, a widely disseminated halophilic bacteria in seafood and brackish water environments, is responsible for wound infections, septicemia, and gastroenteritis (Alipour *et al.* 2014). In our study, we isolated *V. parahaemolyticus* from marine fishes collected from local fish markets in Bangladesh over the year. The result showed that 63.75% (51/80) of marine fishes in this study were positive for *V. parahaemolyticus*. The high

Table 4 Serotype distribution of the *V. parahaemolyticus* isolates from marine fishes

Serotype	Number of Isolates	Percent
O5:KUT	19	26.76
O10:KUT	10	14.09
O2:KUT	7	9.86
O1:KUT	5	7.05
O11:KUT	5	7.05
O4:KUT	2	2.82
O2:K25	1	1.40
O3:K7	1	1.40
O4:K49	1	1.40
O5:K17	1	1.40
O5:K32	1	1.40
O6:K3	1	1.40
O9:KUT	1	1.40
OUT:K53	1	1.40
Untypable	15	21.13
Total	71	100

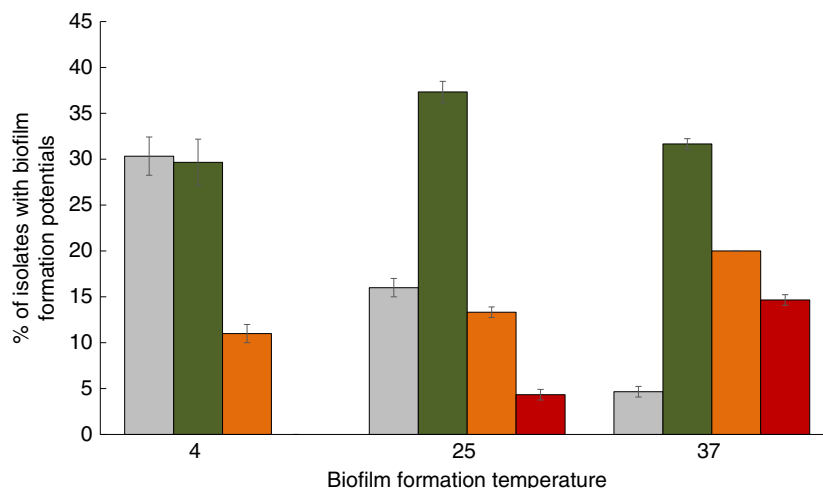
prevalence of *V. parahaemolyticus* in marine fishes is in concordance with the earlier findings (Dileep *et al.* 2003; Ward & Bej 2006; Pal & Das 2010). The greater incidence rate of *V. parahaemolyticus* might be attributed to the gradual increase of environmental temperature that influences the multiplication and perpetuation of the organisms in the environment (Fuenzalida *et al.* 2006). High contamination of marine fishes may permit other transmission routes to humans through contamination of freshwater fishes and secondary contamination of other foods in the kitchen.

The outcome of our study showed the seasonal change of *V. parahaemolyticus* occurrence in marine fishes throughout the year. The higher presence of *V.*

parahaemolyticus was observed during warmer months (April–October) whereas, during colder months their occurrence were gradually decreased (Kaneko & Colwell 1977). Due to the seasonal variation, the water temperature affects the number of *V. parahaemolyticus* in aquaculture (Parveen *et al.* 2008). A positive relation between water temperature and *V. parahaemolyticus* contamination in seafood has been documented in previous studies (DePaola *et al.* 2003; Lhafi & Kühne, 2007; Mahmud *et al.* 2007). *V. parahaemolyticus* overwinter in underneath sediments and re-enter the water body during temperature rises, explaining the process of their seasonal change in the environment (Pfeffer *et al.* 2003).

V. parahaemolyticus harbouring *tdh* and/or *trh* virulent markers pose a significant risk to human health (Ceccarelli *et al.* 2013; Letchumanan *et al.* 2014; Raghunath 2015). Surprisingly, both the *trh* and *tdh* genes were negative for all 71 isolates found in marine fishes. These findings are consistent with previous studies (Parveen *et al.* 2008; Raghunath *et al.* 2008). The complete pathogenesis of *V. parahaemolyticus* is still ambiguous (Ceccarelli *et al.* 2013); while TDH and TRH have been found to be responsible for pathogenesis. *V. parahaemolyticus* still remains pathogenic even when *tdh* or *trh* genes are absent. Some environmental isolates that do not harbour *tdh* and/or *trh* genes can generate other virulence factors like extracellular proteases, biofilm, and siderophore (Mahoney *et al.* 2010; Raghunath 2015).

In our study, we found that most of our isolates were resistant to commonly prescribed antibiotics. The highest resistance pattern (100%) was found against ampicillin and the result is in agreement with some previous studies on seafood (Okuda *et al.* 1997; Han *et al.* 2007; Al-Othubi *et al.* 2014). This resistance profile might be attributed to the misuse of first-generation antibiotics,

Figure 2 Biofilm formation potentiality of *V. parahaemolyticus* at three separate temperatures. Data were represented as the mean $\pm$ SD of three independent experiments. (□) Non-biofilm; (■) weak; (■) moderate; (■) Strong.

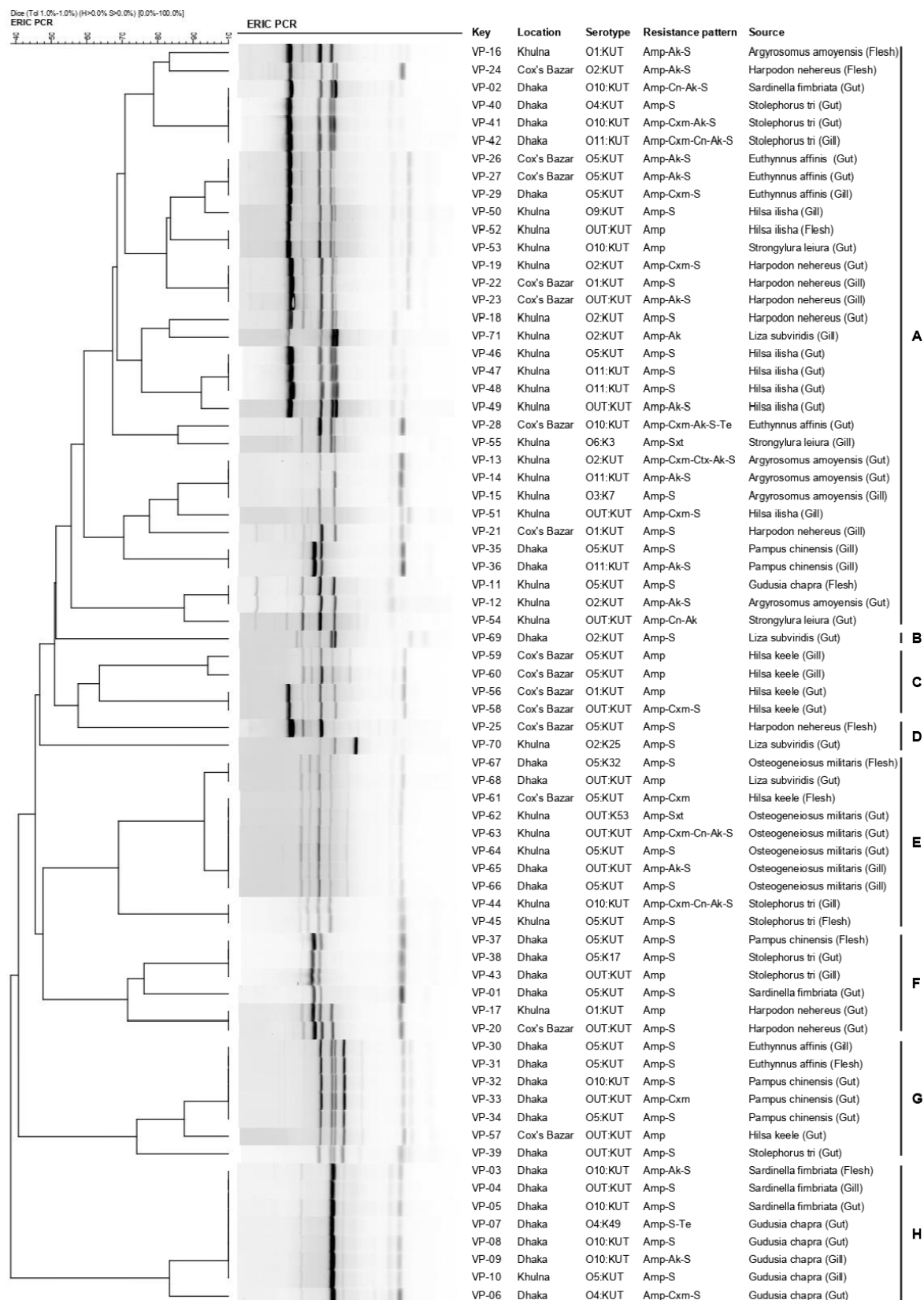


Figure 3 ERIC-PCR Finger Print Analysis of *V. parahaemolyticus* isolates in marine fishes of Bangladesh.

especially ampicillin (Al-Othrubí *et al.* 2014). Other antibiotics to which *V. parahaemolyticus* is resistant are streptomycin (78.9%), amikacin (29.6%) and cefuroxime (18.3%). Previous studies have reported that resistance of streptomycin is common in *V. parahaemolyticus* (Pazhani *et al.* 2014). In the current study, high susceptibility to antibiotics meropenem (100%), chloramphenicol (100%), imipenem (97.2%), nalidixic acid (88.7%), piperacillin/tazobactam (85.9%), trimethoprim-sulfamethoxazole (81.7%), cefotaxime (80.3%), ciprofloxacin (77.5%), and gentamicin (70.4%) were found among our isolates. These outcomes are concordant with previous reports where the majority of the *V. parahaemolyticus* were sensitive to tetracycline and chloramphenicol (Han *et al.* 2007; Sahilah *et al.* 2014; Sudha *et al.* 2014). These antibiotics are still effective against *V. parahaemolyticus* and they could be recommended by physicians to treat *V. parahaemolyticus* associated infections. In this research, 36.7% of isolates have greater than 0.2 MAR value suggesting that fishes came from a highly contaminated origin where multiple antibiotics were used (Krumperman 1983). Nowadays, antibiotics are widely used in clinical, agricultural, aquaculture, poultry and dairy production and because of that antimicrobial residues are spreading in the aquatic environment. These antimicrobial substances present in the environment produce selective pressure on marine bacteria resulting in the increase of MDR bacterial strains in marine life.

V. parahaemolyticus can develop a biofilm that provides a better chance to withstand antibiotics and disinfectants than planktonic cells (Han *et al.* 2016). The production of biofilm is associated with the nature of the bacteria and the surrounding temperature. In this study, we observed that the temperature had an effect on the capacity of *V. parahaemolyticus* to develop a biofilm. The temperature of incubation at 25°C and 37°C increased the formation of biofilms considerably than that of 4°C. Thus the market time-temperature abuse may contribute to the spread and growth of bacteria to harmful levels and the expression of various determinants associated with virulence, including biofilms (Robert-Pillot *et al.* 2004; Mahoney *et al.* 2010). By influencing flagellar motility, temperature plays a crucial role in biofilm formation, which increases the motion of bacteria towards the biofilm surface (Bonsaglia *et al.* 2014). Thus in markets where temperature rises due to ambient temperature, *V. parahaemolyticus* can develop biofilm, leading to a potential threats to consumers. During the long chain of processing and distribution throughout the country, cross-contamination may occur at any point. Slowing down the biofilm formation at lower temperatures ($\leq 4^{\circ}\text{C}$) emphasizes the necessity of careful safety measures to keep seafood under cooling conditions.

To understand the epidemiological diversity, all the isolates were serotyped and the most frequent serogroup was found O5 (30%) (Table 4). Most of the *V. parahaemolyticus* isolates in this study were untypable based on K antigen, a phenomenon reported previously (Nair *et al.* 1985). The high rate of recombination due to mutation, horizontal gene transfer, etc. in the gene for the capsular K antigen than that of somatic O antigen is responsible for this phenomenon. A total of 14 different serotypes was identified from the isolates and 19 with O5:KUT as the dominant serotype, which indicates that a diverse group of *V. parahaemolyticus* are present in marine fishes. Few research group found earlier the geographical confinement of some serogroups (Abbott *et al.* 1989; Osawa *et al.* 1996). Over the last five years in Asia and the United States, the serotypes O3:K6, O4:K68, and O1:KUT (K untypeable) were responsible mainly for *V. parahaemolyticus* associated infections (Chiou *et al.* 2000; Daniels *et al.* 2000). Our results indicate that pandemic traits O1:KUT was present in five *V. parahaemolyticus* isolates. Though the occurrence of environmental pandemic strains is scanty, the possible risk of outbreaks should be critically monitored.

ERIC-PCR is usually considered for the analysis of genetic diversity of *V. parahaemolyticus*, as it has high discriminatory power, reliability and rapid typing ability (Khan *et al.* 2002). This technique is reasonably straightforward as well as cost-effective to study a large number of strains. Using this method, we classified the isolates into eight clusters, at 55% similarity. Most of the isolates were distributed in cluster A which may indicate that there are some genetic associations among those strains but these strains are originated from a different location and from different fish samples. In cluster C, all the strains are originated from *Hilsa kelee* fish collected from Cox's Bazar, which imply their genetic relatedness. In cluster H, most of the strains collected from Dhaka have the same molecular pattern, though they were originated from two different fishes. This may imply the dissemination of *V. parahaemolyticus* in different fishes during the time of transport or distribution in the fish markets. Clustering based on ERIC-PCR was not dependent on the isolation location or profile of antibiotic resistance or serotyping. Our results are comparable with previous findings, which confirm the genetic diversity within *V. parahaemolyticus* strains (Khan *et al.* 2002; Xie *et al.* 2016). The high genetic diversity of *V. parahaemolyticus* could be attributed to the adaptive changes in the bacterium genome while persisting in the environment and during infection (Wong *et al.* 2000).

The limitation of our study is that we could not cover an extensive sampling area throughout the country as

well as the sufficient sample variation and number due to time restrictions. Rigorous sampling, as well as genomic analysis of *V. parahaemolyticus* from marine fishes and other seafood, might clarify their genotypic and phenotypic diversity in Bangladesh. However, this study confirms the diversity of *V. parahaemolyticus* in marine fishes collected from different fish markets in Bangladesh. Our study provides comprehensive research on *V. parahaemolyticus* isolated from marine fishes by highlighting the prevalence, antibiotic resistance pattern, serotyping, biofilm formation and genetic diversity. Their resistance to multiple antibiotics could be a threat to public health should not be overlooked. The clustering of the isolates shows their genetic diversity. The biofilm formation capacity indicates the persistence of these organisms at different temperatures. The presence of different serotypes including few pandemic serotypes might indicate the diversity and outbreak potential of these isolates. Health risk may arise through cross-contamination by marine fishes to freshwater fishes in the market. Most of these concerns could be fixed by maintaining proper cleanliness at the fish market. Marine fishes are becoming a common and popular protein source throughout the country due to easy transportation. To confront the possible risk implications of *V. parahaemolyticus*, continuous monitoring is crucial to confirm the safety of these food products.

Author Contributions

SA, ZHM, MSI, MAA and MMR planned and organized the study. ZHM, MSI, SA, MH, and MM were involved in implementing the study. SA, ABA, ABS, MAA and MH carried out laboratory work. ZHM, SA, MH, MM, MSI, MBA and DM were involved in interpreting the data. SA, MH, ZHM and MM had a major contribution in writing the manuscript. All authors contributed in proofreading the manuscript.

Acknowledgments

This study was funded by core donors who provide unrestricted support to the institutional fund of the International Centre for Diarrhoeal Disease Research, Bangladesh (icddr,b) for its operations and research. Current donors providing unrestricted support include the Governments of Bangladesh, Canada, Sweden, and the UK. We gratefully acknowledge our core donors for their support and commitment to icddr,b's research efforts.

Conflict of Interest

The authors declare no conflict of interest.

References

- Abbott, S.L., Powers, C., Kaysner, C.A., Takeda, Y., Ishibashi, M., Joseph, S.W. and Janda, J.M. (1989) Emergence of a restricted bioserovar of *Vibrio parahaemolyticus* as the predominant cause of *Vibrio*-associated gastroenteritis on the West Coast of the United States and Mexico. *J Clin Microbiol* **27**, 2891–2893.
- Alam, M., Chowdhury, W.B., Bhuiyan, N.A., Islam, A., Hasan, N.A., Nair, G.B., Watanabe, H., Siddique, A.K. et al. (2009) Serogroup, virulence, and genetic traits of *Vibrio parahaemolyticus* in the estuarine ecosystem of Bangladesh. *Appl Environ Microbiol* **75**, 6268–6274.
- Alipour, M., Issazadeh, K. and Soleimani, J. (2014) Isolation and identification of *Vibrio parahaemolyticus* from seawater and sediment samples in the southern coast of the Caspian Sea. *Comp Clin Path* **23**, 129–133.
- Al-Othubi, S.M.Y., Kqueen, C.Y., Mirhosseini, H., Hadi, Y.A. and Radu, S. (2014) Antibiotic resistance of *Vibrio parahaemolyticus* isolated from cockles and shrimp sea food marketed in Selangor, Malaysia. *Clin Microbiol* **3**, 148.
- Baffone, W., Tarsi, R., Pane, L., Campana, R., Repetto, B., Mariottini, G.L. and Pruzzo, C. (2006) Detection of free-living and plankton-bound vibrios in coastal waters of the Adriatic Sea (Italy) and study of their pathogenicity-associated properties. *Environ Microbiol* **8**, 1299–1305.
- Bej, A.K., Patterson, D.P., Brasher, C.W., Vickery, M.C.L., Jones, D.D. and Kaysner, C.A. (1999) Detection of total and hemolysin-producing *Vibrio parahaemolyticus* in shellfish using multiplex PCR amplification of *tlh*, *tdh* and *trh*. *J Microbiol Methods* **36**, 215–225.
- Bhuiyan, N.A., Ansaruzzaman, M., Kamruzzaman, M., Alam, K., Chowdhury, N.R., Nishibuchi, M., Faruque, S.M., Sack, D.A. et al. (2002) Prevalence of the pandemic genotype of *Vibrio parahaemolyticus* in Dhaka, Bangladesh, and significance of its distribution across different serotypes. *J Clin Microbiol* **40**, 284–286.
- Bonsaglia, E.C.R., Silva, N.C.C., Júnior, A.F., Júnior, J.P.A., Tsunemi, M.H. and Rall, V.L.M. (2014) Production of biofilm by *Listeria monocytogenes* in different materials and temperatures. *Food Control* **35**, 386–391.
- Ceccarelli, D., Hasan, N.A., Huq, A. and Colwell, R.R. (2013) Distribution and dynamics of epidemic and pandemic *Vibrio parahaemolyticus* virulence factors. *Front Cell Infect Microbiol* **3**, 97.
- Chen, W., Xie, Y., Xu, J., Wang, Q., Gu, M., Yang, J., Zhou, M., Wang, D. et al. (2012) Molecular typing of *Vibrio parahaemolyticus* isolates from the middle-east coastline of China. *Int J Food Microbiol* **153**, 402–412.
- Chiou, C.-S., Hsu, S.-Y., Chiu, S.-I., Wang, T.-K. and Chao, C.-S. (2000) *Vibrio parahaemolyticus* serovar O3: K6 as cause of unusually high incidence of food-borne disease outbreaks in Taiwan from 1996 to 1999. *J Clin Microbiol* **38**, 4621–4625.

- Daniels, N.A., MacKinnon, L., Bishop, R., Altekruze, S., Ray, B., Hammond, R.M., Thompson, S., Wilson, S. *et al.* (2000) *Vibrio parahaemolyticus* infections in the United States, 1973–1998. *J Infect Dis* **181**, 1661–1666.
- DePaola, A., Nordstrom, J.L., Bowers, J.C., Wells, J.G. and Cook, D.W. (2003) Seasonal abundance of total and pathogenic *Vibrio parahaemolyticus* in Alabama oysters. *Appl Environ Microbiol* **69**, 1521–1526.
- Dileep, V., Kumar, H.S., Kumar, Y., Nishibuchi, M., Karunasagar, I. and Karunasagar, I. (2003) Application of polymerase chain reaction for detection of *Vibrio parahaemolyticus* associated with tropical seafoods and coastal environment. *Lett Appl Microbiol* **36**, 423–427.
- Drake, S.L., DePaola, A. and Jaykus, L. (2007) An overview of *Vibrio vulnificus* and *Vibrio parahaemolyticus*. *Compr Rev Food Sci Food Saf* **6**, 120–144.
- Duran, G.M. and Marshall, D.L. (2005) Ready-to-eat shrimp as an international vehicle of antibiotic-resistant bacteria. *J Food Prot* **68**, 2395–2401.
- Fuenzalida, L., Hernández, C., Toro, J., Rioseco, M.L., Romero, J. and Espejo, R.T. (2006) *Vibrio parahaemolyticus* in shellfish and clinical samples during two large epidemics of diarrhoea in southern Chile. *Environ Microbiol* **8**, 675–683.
- Guglielmetti, E., Korhonen, J.M., Heikkinen, J., Morelli, L. and Von Wright, A. (2009) Transfer of plasmid-mediated resistance to tetracycline in pathogenic bacteria from fish and aquaculture environments. *FEMS Microbiol Lett* **293**, 28–34.
- Han, F., Walker, R.D., Janes, M.E., Prinyawiwatkul, W. and Ge, B. (2007) Antimicrobial susceptibilities of *Vibrio parahaemolyticus* and *Vibrio vulnificus* isolates from Louisiana Gulf and retail raw oysters. *Appl Environ Microbiol* **73**, 7096–7098.
- Han, N., Mizan, M.F.R., Jahid, I.K. and Ha, S.-D. (2016) Biofilm formation by *Vibrio parahaemolyticus* on food and food contact surfaces increases with rise in temperature. *Food Control* **70**, 161–166.
- Hara-Kudo, Y., Nishina, T., Nakagawa, H., Konuma, H., Hasegawa, J. and Kumagai, S. (2001) Improved method for detection of *Vibrio parahaemolyticus* in seafood. *Appl Environ Microbiol* **67**, 5819–5823.
- Hassan, A., Usman, J., Kaleem, F., Omair, M., Khalid, A. and Iqbal, M. (2011) Evaluation of different detection methods of biofilm formation in the clinical isolates. *Brazilian J Infect Dis* **15**, 305–311.
- Hindler, J.A. and Jorgensen, J.H. (2011) Susceptibility test methods: fastidious bacteria. In *Manual of Clinical Microbiology*, 10th edn pp. 1180–1203. Washington DC: American Society of Microbiology.
- Honda, T., Iida, T., Akeda, Y. and Kadama, T. (2008) Sixty years of *Vibrio parahaemolyticus* Research-Despite decades of research following a severe foodborne outbreak in Japan, this pathogen remains mysterious at the molecular level. *Microbe* **3**, 462.
- Honda, T., Ni, Y., Yoh, M. and Miwatani, T. (1989) Production of monoclonal antibodies against thermostable direct hemolysin of *Vibrio parahaemolyticus* and application of the antibodies for enzyme-linked immunosorbent assay. *Med Microbiol Immunol* **178**, 245–253.
- Islam, M.S., Tasmin, R., Khan, S.I., Bakht, H.B.M., Mahmood, Z.H., Rahman, M.Z., Bhuiyan, N.A., Nishibuchi, M. *et al.* (2004) Pandemic strains of O3: K6 *Vibrio parahaemolyticus* in the aquatic environment of Bangladesh. *Can J Microbiol* **50**, 827–834.
- Jerbi, M.A., Ouanes, Z., Besbes, R., Achour, L. and Kacem, A. (2011) Single and combined genotoxic and cytotoxic effects of two xenobiotics widely used in intensive aquaculture. *Mutat Res Toxicol Environ Mutagen* **724**, 22–27.
- Joseph, S.W., Colwell, R.R. and Kaper, J.B. (1982) *Vibrio parahaemolyticus* and related halophilic vibrios. *CRC Crit Rev Microbiol* **10**, 77–124.
- Kaneko, T. and Colwell, R.R. (1977) The annual cycle of *Vibrio parahaemolyticus* in Chesapeake Bay. *Microb Ecol* **4**, 135–155.
- Kaysner, C.A. and DePaola, A. Jr (2004) *Vibrio cholerae*, *V. parahaemolyticus*, *V. vulnificus*, and Other *Vibrio* spp. Bacteriological Analytical Manual.
- Khan, A.A., McCarthy, S., Wang, R.-F. and Cerniglia, C.E. (2002) Characterization of United States outbreak isolates of *Vibrio parahaemolyticus* using enterobacterial repetitive intergenic consensus (ERIC) PCR and development of a rapid PCR method for detection of O3: K6 isolates. *FEMS Microbiol Lett* **206**, 209–214.
- Krumperman, P.H. (1983) Multiple antibiotic resistance indexing of *Escherichia coli* to identify high-risk sources of fecal contamination of foods. *Appl Environ Microbiol* **46**, 165–170.
- Letchumanan, V., Chan, K.-G. and Lee, L.-H. (2014) *Vibrio parahaemolyticus*: a review on the pathogenesis, prevalence, and advance molecular identification techniques. *Front Microbiol* **5**, 705.
- Lhafi, S.K. and Kühne, M. (2007) Occurrence of *Vibrio* spp. in blue mussels (*Mytilus edulis*) from the German Wadden Sea. *Int J Food Microbiol* **116**, 297–300.
- Mahmud, Z.H., Kasso, A., Mohammad, A., Yamato, M., Bhuiyan, N.A., Nair, G.B. and Ota, F. (2006) Isolation and molecular characterization of toxigenic *Vibrio parahaemolyticus* from the Kii channel, Japan. *Microbiol Res* **161**, 25–37.
- Mahmud, Z.H., Neogi, S.B., Kasso, A., Wada, T., Islam, M.S., Nair, G.B. and Ota, F. (2007) Seaweeds as a reservoir for diverse *Vibrio parahaemolyticus* populations in Japan. *Int J Food Microbiol* **118**, 92–96.
- Mahoney, J.C., Gerding, M.J., Jones, S.H. and Whistler, C.A. (2010) Comparison of the pathogenic potentials of environmental and clinical *Vibrio parahaemolyticus* strains indicates a role for temperature regulation in virulence. *Appl Environ Microbiol* **76**, 7459–7465.

- Marshall, S., Clark, C.G., Wang, G., Mulvey, M., Kelly, M.T. and Johnson, W.M. (1999) Comparison of molecular methods for typing *Vibrio parahaemolyticus*. *J Clin Microbiol* **37**, 2473–2478.
- Martinez-Urtaza, J., Simental, L., Velasco, D., DePaola, A., Ishibashi, M., Nakaguchi, Y., Nishibuchi, M., Carrera-Flores, D. et al. (2005) Pandemic *Vibrio parahaemolyticus* O3: K6, Europe. *Emerg Infect Dis* **11**, 1319.
- Matsuda, S., Kodama, T., Okada, N., Okayama, K., Honda, T. and Iida, T. (2010) Association of *Vibrio parahaemolyticus* thermostable direct hemolysin with lipid rafts is essential for cytotoxicity but not hemolytic activity. *Infect Immun* **78**, 603–610.
- Nair, G.B., Sarkar, B.L., Abraham, M. and Pal, S.C. (1985) Serotypes of *Vibrio parahaemolyticus* isolates from hydrobiologically dissimilar aquatic environments. *Appl Environ Microbiol* **50**, 724–726.
- O'Toole, G.A. (2011) Microtiter dish biofilm formation assay. *J Vis Exp* **47**, e2437.
- Okuda, J., Ishibashi, M., Hayakawa, E., Nishino, T., Takeda, Y., Mukhopadhyay, A.K., Garg, S., Bhattacharya, S.K. et al. (1997) Emergence of a unique O3: K6 clone of *Vibrio parahaemolyticus* in Calcutta, India, and isolation of strains from the same clonal group from Southeast Asian travelers arriving in Japan. *J Clin Microbiol* **35**, 3150–3155.
- Osawa, R., Okitsu, T., Morozumi, H. and Yamai, S. (1996) Occurrence of urease-positive *Vibrio parahaemolyticus* in Kanagawa, Japan, with specific reference to presence of thermostable direct hemolysin (TDH) and the TDH-related-hemolysin genes. *Appl Environ Microbiol* **62**, 725–727.
- Pal, D. and Das, N. (2010) Isolation, identification and molecular characterization of *Vibrio parahaemolyticus* from fish samples in Kolkata. *Eur Rev Med Pharmacol Sci* **14**, 545–549.
- Parveen, S., Hettiarachchi, K.A., Bowers, J.C., Jones, J.L., Tamplin, M.L., McKay, R., Beatty, W., Brohawn, K. et al. (2008) Seasonal distribution of total and pathogenic *Vibrio parahaemolyticus* in Chesapeake Bay oysters and waters. *Int J Food Microbiol* **128**, 354–361.
- Patel, J.B., Cockerill, F.R. and Bradford, P.A. (2015) Performance standards for antimicrobial susceptibility testing: twenty-fifth informational supplement. Clinical and Laboratory Standards Institute.
- Pazhani, G.P., Bhowmik, S.K., Ghosh, S., Guin, S., Dutta, S., Rajendran, K., Saha, D.R., Nandy, R.K. et al. (2014) Trends in the epidemiology of pandemic and non-pandemic strains of *Vibrio parahaemolyticus* isolated from diarrheal patients in Kolkata, India. *PLoS Negl Trop Dis* **8**, e2815.
- Pfeffer, C.S., Hite, M.F. and Oliver, J.D. (2003) Ecology of *Vibrio vulnificus* in estuarine waters of eastern North Carolina. *Appl Environ Microbiol* **69**, 3526–3531.
- Qadri, F., Chowdhury, N.R., Takeda, Y. and Nair, G.B. (2005) *Vibrio parahaemolyticus*—seafood safety and associations with higher organisms. *Oceans and health: pathogens in the marine environment*. pp. 277–295. Boston, MA: Springer.
- Quilici, M.-L., Robert-Pillot, A., Picart, J. and Fournier, J.-M. (2005) Pandemic *Vibrio parahaemolyticus* O3: K6 spread, France. *Emerg Infect Dis* **11**, 1148.
- Raghunath, P. (2015) Roles of thermostable direct hemolysin (TDH) and TDH-related hemolysin (TRH) in *Vibrio parahaemolyticus*. *Front Microbiol* **5**, 805.
- Raghunath, P., Acharya, S., Bhanumathi, A., Karunasagar, I. and Karunasagar, I. (2008) Detection and molecular characterization of *Vibrio parahaemolyticus* isolated from seafood harvested along the southwest coast of India. *Food Microbiol* **25**, 824–830.
- Rivera, I.G., Chowdhury, M.A., Huq, A., Jacobs, D., Martins, M.T. and Colwell, R.R. (1995) Enterobacterial repetitive intergenic consensus sequences and the PCR to generate fingerprints of genomic DNAs from *Vibrio cholerae* O1, O139, and non-O1 strains. *Appl Environ Microbiol* **61**, 2898–2904.
- Robert-Pillot, A., Guénolé, A., Lesne, J., Delesmont, R., Fournier, J.-M. and Quilici, M.-L. (2004) Occurrence of the *tdh* and *trh* genes in *Vibrio parahaemolyticus* isolates from waters and raw shellfish collected in two French coastal areas and from seafood imported into France. *Int J Food Microbiol* **91**, 319–325.
- Sahilah, A.M., Laila, R.A.S., Sallehuddin, H.M., Osman, H., Aminah, A. and Azuhairi, A.A. (2014) Antibiotic resistance and molecular typing among cockle (*Anadara granosa*) strains of *Vibrio parahaemolyticus* by polymerase chain reaction (PCR)-based analysis. *World J Microbiol Biotechnol* **30**, 649–659.
- Siddique, A.B., Moniruzzaman, M., Ali, S., Dewan, M.N., Islam, M.R., Islam, M.S., Amin, M.B., Mondal, D. et al. (2021) Characterization of pathogenic *Vibrio parahaemolyticus* isolated from fish aquaculture of the south-west coastal area of Bangladesh. *Front Microbiol* **12**, 635539.
- Sudha, S., Mridula, C., Silvester, R. and Hatha, A.A.M. (2014) Prevalence and antibiotic resistance of pathogenic Vibrios in shellfishes from Cochin market. *Indian J Geo-Mar Sci* **43**, 815–824.
- Tang, G.-Q., Lida, T., Yamamoto, K. and Honda, T. (1995) Ca²⁺-independent cytotoxicity of *Vibrio parahaemolyticus* thermostable direct hemolysin (TDH) on Intestine 407, a cell line derived from human embryonic intestine. *FEMS Microbiol Lett* **134**, 233–238.
- Tendencia, E.A. and de la Peña, L.D. (2001) Antibiotic resistance of bacteria from shrimp ponds. *Aquaculture* **195**, 193–204.
- Versalovic, J., Koeuth, T. and Lupski, R. (1991) Distribution of repetitive DNA sequences in eubacteria and application to fingerprinting of bacterial genomes. *Nucleic Acids Res.* **19**, 6823–6831.
- Wang, R., Zhong, Y., Gu, X., Yuan, J., Saeed, A.F. and Wang, S. (2015) The pathogenesis, detection, and prevention of *Vibrio parahaemolyticus*. *Front Microbiol* **6**, 144.

- Ward, L.N. and Bej, A.K. (2006) Detection of *Vibrio parahaemolyticus* in shellfish by use of multiplexed real-time PCR with TaqMan fluorescent probes. *Appl Environ Microbiol* **72**, 2031–2042.
- Wong, H.-C., Liu, S.-H., Ku, L.-W., Lee, I.-Y., Wang, T.-K., Lee, Y.-S., Lee, C.-L., Kuo, L.-P. *et al.* (2000) Characterization of *Vibrio parahaemolyticus* isolates obtained from foodborne illness outbreaks during 1992 through 1995 in Taiwan. *J Food Prot* **63**, 900–906.
- Xie, T., Xu, X., Wu, Q., Zhang, J. and Cheng, J. (2016) Prevalence, molecular characterization, and antibiotic susceptibility of *Vibrio parahaemolyticus* from ready-to-eat foods in China. *Front Microbiol* **7**, 549.

Supporting Information

Additional Supporting Information may be found in the online version of this article:

Table S1 Distribution of fish collection throughout the year.

Table S2 Prevalence of *Vibrio parahaemolyticus* in different parts of marine fishes.

Table S3 The average value inhibition zone (mm) of different isolates against different antibiotics.

Table S4 Zone diameter interpretive standards for *Vibrio parahaemolyticus*.